AIML Syllabus complete

You said:

complete AIML syllabus basic to advanced chapterwise each topic mentioned detailed

ChatGPT said:

Complete AI/ML Syllabus: Basic to Advanced

This syllabus follows the correct learning order for becoming an **AI/ML Engineer**. It covers programming, mathematics, data analysis, machine learning, deep learning, NLP, computer vision, Generative AI, RAG, deployment, MLOps, projects, and interviews.

Phase 1: AI and Machine Learning Foundations

Chapter 1: Introduction to Artificial Intelligence

Topics

- Definition of Artificial Intelligence
- History and evolution of AI
- Goals of AI
- AI vs ML vs Deep Learning vs Data Science
- Intelligent agents
- Environment, state, action and reward
- Rational agents
- Types of environments:
  - Fully vs partially observable
  - Deterministic vs stochastic
  - Static vs dynamic
  - Discrete vs continuous
  - Single-agent vs multi-agent
- Applications of AI
- Advantages and limitations
- Responsible and ethical AI

Types of AI by capability

- Artificial Narrow Intelligence
- Artificial General Intelligence

- Artificial Super Intelligence

Types of AI by functionality

- Reactive machines
- Limited-memory AI
- Theory-of-mind AI
- Self-aware AI

Chapter 2: Introduction to Machine Learning

Topics

- Definition of machine learning
- How machines learn from data
- Traditional programming vs machine learning
- Dataset, observation and sample
- Features and target variables
- Training and inference
- Model, algorithm and prediction
- Parameters and hyperparameters
- Loss function and cost function
- Generalization
- Bias and variance
- Underfitting and overfitting

Types of machine learning

1. Supervised learning
2. Unsupervised learning
3. Semi-supervised learning
4. Self-supervised learning
5. Reinforcement learning
6. Online learning
7. Batch learning
8. Transfer learning
9. Active learning

Common ML tasks

- Classification
- Regression
- Clustering
- Dimensionality reduction

- Anomaly detection
- Recommendation
- Ranking
- Forecasting

Chapter 3: Complete ML Lifecycle

Workflow

```
Problem definition
      ↓
Data collection
      ↓
Data understanding
      ↓
Data cleaning
      ↓
Exploratory analysis
      ↓
Feature engineering
      ↓
Data splitting
      ↓
Model selection
      ↓
Model training
      ↓
Model evaluation
      ↓
Hyperparameter tuning
      ↓
Deployment
      ↓
Monitoring and retraining
```

Topics

- Business problem understanding
- Translating business problems into ML problems
- Defining input, output and success metrics
- Baseline model
- Experiment tracking
- Offline and online evaluation
- Deployment
- Data drift

- Concept drift
- Model retraining

Phase 2: Python Programming

Chapter 4: Python Fundamentals

Topics

- Installing Python
- Python interpreter
- Jupyter Notebook
- Google Colab
- VS Code
- Python syntax and indentation
- Comments
- Keywords and identifiers
- Variables
- Naming conventions
- Dynamic typing
- Multiple assignment
- Constants
- type() and id()

Data types

- Integer
- Float
- Complex
- Boolean
- String
- List
- Tuple
- Set
- Dictionary
- Range
- Bytes
- None

Type conversion

- Implicit conversion
- Explicit conversion

- int()
- float()
- str()
- bool()
- list()
- tuple()
- set()

Chapter 5: Operators and Input/Output

Operators

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Identity operators
- Membership operators
- Bitwise operators
- Operator precedence
- Associativity

Input and output

- input()
- print()
- sep
- end
- Formatted strings
- f-strings
- format()
- Escape characters

Chapter 6: Conditions and Loops

Conditional statements

- if
- if-else
- if-elif-else
- Nested conditions
- Ternary operator

- match-case
- Truthy and falsy values

Loops

- for
- while
- Nested loops
- range()
- break
- continue
- pass
- Loop else
- enumerate()
- zip()

Pattern problems

- Number patterns
- Star patterns
- Pyramid patterns
- Searching
- Counting
- Prime-number problems
- Fibonacci series
- Factorial

Chapter 7: Strings

Topics

- String creation
- Indexing
- Positive and negative indexing
- Slicing
- String immutability
- Traversing strings
- String comparison
- Concatenation and repetition
- Membership testing

- Unicode

Important methods

- upper() and lower()
- title() and capitalize()
- strip(), lstrip(), rstrip()
- split() and join()
- find() and index()
- replace()
- count()
- startswith() and endswith()
- isalpha(), isdigit(), isalnum()
- partition()

Interview problems

- Reverse a string
- Palindrome
- Anagram
- Character frequency
- First non-repeating character
- Longest substring
- Remove duplicate characters
- String compression

Chapter 8: Lists and Tuples

Lists

- Creating lists
- Indexing and slicing
- Updating elements
- Adding and removing elements
- Nested lists
- Shallow and deep copying
- List comprehension

List methods

- append()
- extend()
- insert()
- remove()

- pop()
- clear()
- index()
- count()
- sort()
- reverse()
- copy()

Tuples

- Creating tuples
- Single-element tuple
- Packing and unpacking
- Extended unpacking
- Tuple methods
- Tuple immutability
- List vs tuple
- Named tuples

Interview problems

- Find maximum and minimum
- Remove duplicates
- Second-largest number
- Rotate a list
- Merge sorted lists
- Flatten nested lists
- Frequency counting
- Pair sum

Chapter 9: Sets and Dictionaries

Sets

- Creating sets
- Unique values
- Mutable sets
- frozenset
- Union
- Intersection
- Difference
- Symmetric difference
- Subset and superset

- Set comprehension

Set methods

- add()
- update()
- remove()
- discard()
- pop()
- clear()

Dictionaries

- Key-value structure
- Creating and updating dictionaries
- Accessing values
- Nested dictionaries
- Dictionary comprehension
- Dictionary merging
- Hashable keys

Dictionary methods

- get()
- keys()
- values()
- items()
- update()
- pop()
- popitem()
- setdefault()
- fromkeys()

Chapter 10: Functions

Topics

- Defining and calling functions
- Function parameters and arguments
- Positional arguments
- Keyword arguments
- Default arguments
- Variable-length arguments
- *args

- **kwargs
- Return values
- Multiple return values
- Local and global scope
- global
- nonlocal
- Lambda functions
- Recursion
- First-class functions
- Higher-order functions
- Closures
- Decorators

Functional programming

- map()
- filter()
- reduce()
- any()
- all()
- sorted() with key functions

Chapter 11: Object-Oriented Programming

Core concepts

- Class and object
- Attributes and methods
- Constructor
- self
- Instance variables
- Class variables
- Instance methods
- Class methods
- Static methods

Four pillars

1. Encapsulation
2. Inheritance
3. Polymorphism

4. Abstraction

Advanced OOP

- Single inheritance
- Multiple inheritance
- Multilevel inheritance
- Hierarchical inheritance
- Method overriding
- Method overloading alternatives
- Operator overloading
- Abstract classes
- Interfaces through abstract base classes
- super()
- Method Resolution Order
- Composition
- Aggregation
- Magic methods
- Properties
- Getters and setters
- Dataclasses

Chapter 12: Exception and File Handling

Exceptions

- Syntax errors
- Runtime errors
- Logical errors
- try
- except
- else
- finally
- raise
- Custom exceptions
- Exception hierarchy

File handling

- Text and binary files
- File modes
- Reading and writing
- with statement

- File pointer
- CSV files
- JSON files
- Pickle files
- Path handling

Chapter 13: Advanced Python

Topics

- Modules and packages
- Virtual environments
- pip
- Iterators
- Generators
- Generator expressions
- Decorators
- Regular expressions
- Context managers
- Type hints
- Logging
- Unit testing
- Assertions
- Shallow and deep copy
- Memory management
- Garbage collection
- Multithreading
- Multiprocessing
- Asynchronous programming
- async and await
- API requests
- Web scraping basics

Phase 3: Data Structures and Algorithms

Chapter 14: Complexity Analysis

- Time complexity
- Space complexity
- Big O notation
- Best, average and worst cases

- Constant, logarithmic, linear and quadratic complexity
- Recursive complexity
- Amortized analysis

Chapter 15: Core Data Structures

- Arrays
- Linked lists
- Stacks
- Queues
- Hash maps
- Hash sets
- Trees
- Binary search trees
- Heaps
- Graphs
- Tries

Chapter 16: Problem-Solving Patterns

- Two pointers
- Sliding window
- Prefix sum
- Hashing
- Binary search
- Fast and slow pointers
- Monotonic stack
- Recursion
- Backtracking
- Greedy algorithms
- Divide and conquer
- Dynamic programming
- Graph traversal
- BFS and DFS

Phase 4: Mathematics for AI/ML

Chapter 17: Basic Mathematics

- Functions
- Linear and nonlinear functions
- Exponents
- Logarithms

- Summations
- Sequences and series
- Polynomial equations
- Coordinate geometry
- Distance measures

Common distance formulas

Euclidean distance:

$$ d(x,y)=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} $$

Manhattan distance:

$$ d(x,y)=\sum_{i=1}^{n}|x_i-y_i| $$

Minkowski distance:

$$ d(x,y)=\left(\sum_{i=1}^{n}|x_i-y_i|^p\right)^{1/p} $$

Chapter 18: Linear Algebra

Topics

- Scalars
- Vectors
- Matrices
- Tensors
- Matrix shapes
- Vector addition
- Scalar multiplication
- Dot product
- Matrix multiplication
- Transpose
- Identity matrix
- Diagonal matrix
- Symmetric matrix
- Orthogonal matrix
- Matrix inverse
- Determinant
- Rank
- Linear independence
- Basis and dimension
- Eigenvalues and eigenvectors

- Singular Value Decomposition
- Norms
- Cosine similarity
- Matrix factorization

Cosine similarity:

[\cos(\theta)=\frac{A\cdot B}{|A||B|}]

Applications:

- PCA
- Neural networks
- Embeddings
- Recommendation systems
- Image processing

Chapter 19: Calculus

Topics

- Limits
- Continuity
- Derivatives
- Partial derivatives
- Chain rule
- Product rule
- Gradients
- Jacobian matrix
- Hessian matrix
- Maxima and minima
- Convex and non-convex functions
- Integration
- Gradient descent
- Backpropagation

Gradient-descent update:

[\theta_{new}=\theta_{old}-\alpha\nabla J(\theta)]

Where:

- (\theta): model parameters
- (\alpha): learning rate

- (J): cost function
- (\nabla J): gradient

Chapter 20: Probability

Topics

- Random experiments
- Sample spaces
- Events
- Independent and dependent events
- Mutually exclusive events
- Conditional probability
- Joint probability
- Marginal probability
- Bayes' theorem
- Random variables
- Probability mass function
- Probability density function
- Cumulative distribution
- Expectation
- Variance
- Standard deviation
- Covariance

Bayes' theorem:

[P(A|B)=\frac{P(B|A)P(A)}{P(B)}]

Distributions

- Bernoulli
- Binomial
- Normal
- Uniform
- Poisson
- Exponential
- Multinomial
- Gaussian mixture

Chapter 21: Statistics

Descriptive statistics

- Population and sample
- Mean
- Median
- Mode
- Range
- Variance
- Standard deviation
- Quartiles
- Percentiles
- Interquartile range
- Skewness
- Kurtosis
- Covariance
- Correlation

Population variance:

[\sigma^2=\frac{\sum_{i=1}^{N}(x_i-\mu)^2}{N}]

Inferential statistics

- Sampling
- Sampling distributions
- Central Limit Theorem
- Standard error
- Confidence intervals
- Point and interval estimation
- Hypothesis testing
- Null and alternative hypotheses
- Significance level
- p-value
- Type I and Type II errors
- Statistical power

Statistical tests

- Z-test
- One-sample t-test
- Independent t-test
- Paired t-test

- Chi-square test
- ANOVA
- Mann–Whitney U
- Wilcoxon test
- Kruskal–Wallis test

Phase 5: NumPy, Pandas and Visualization

Chapter 22: NumPy

Topics

- NumPy arrays
- Array dimensions and shapes
- Data types
- Array creation
- Indexing and slicing
- Boolean indexing
- Fancy indexing
- Reshaping
- Flattening
- Concatenating
- Splitting
- Broadcasting
- Vectorization
- Aggregation
- Sorting
- Random numbers
- Missing values
- Linear algebra
- Eigenvalues and eigenvectors
- Determinants and inverses
- SVD
- einsum
- Memory optimization

Chapter 23: Pandas

Core topics

- Series
- DataFrame

- Index
- Reading CSV, Excel, JSON and SQL
- Inspecting data
- Selecting rows and columns
- loc and iloc
- Boolean filtering
- query()

Data cleaning

- Missing values
- Duplicate values
- Incorrect data types
- String cleaning
- Datetime handling
- Category standardization
- Outliers

Transformation

- Sorting
- Renaming
- Replacing
- Mapping
- Applying functions
- Binning
- Categorical data

Aggregation

- groupby
- agg
- Pivot tables
- Crosstabs
- Rolling calculations
- Expanding calculations

Combining data

- Merge
- Join
- Concatenate

- Append alternatives

Advanced Pandas

- MultiIndex
- Window functions
- Time-series analysis
- Vectorized operations
- Memory optimization
- Method chaining

Chapter 24: Data Visualization

Matplotlib

- Figure and axes
- Line chart
- Bar chart
- Scatter plot
- Histogram
- Box plot
- Pie chart
- Area chart
- Stack plot
- Error-bar plot
- Heatmap
- Subplots
- Labels, titles and legends
- Annotations
- Styles and themes
- Saving figures

Seaborn

- Distribution plots
- Count plot
- Bar plot
- Box plot
- Violin plot
- Strip plot
- Swarm plot
- Scatter plot
- Regression plot

- Pair plot
- Joint plot
- Heatmap
- Facet grid

Interactive visualization

- Plotly
- Bokeh
- Streamlit charts
- Dashboard design
- Data storytelling
- Choosing the correct graph

Phase 6: SQL and Databases

Chapter 25: Database Fundamentals

- Data and databases
- DBMS and RDBMS
- Tables, rows and columns
- Primary and foreign keys
- Candidate and composite keys
- Constraints
- Relationships
- Normalization
- Denormalization
- ACID properties
- Transactions

Chapter 26: SQL Basics

- CREATE
- ALTER
- DROP
- TRUNCATE
- INSERT
- UPDATE
- DELETE
- SELECT
- WHERE
- ORDER BY
- DISTINCT

- LIMIT
- BETWEEN
- IN
- LIKE
- CASE

Chapter 27: SQL Intermediate and Advanced

- Aggregate functions
- GROUP BY
- HAVING
- Inner join
- Left join
- Right join
- Full join
- Cross join
- Self join
- Subqueries
- Correlated subqueries
- CTE
- Recursive CTE
- Views
- Indexes
- Stored procedures
- Triggers
- Transactions

Window functions

- ROW_NUMBER
- RANK
- DENSE_RANK
- LAG
- LEAD
- FIRST_VALUE
- LAST_VALUE
- Running totals
- Moving averages
- Partitioning

Phase 7: Data Collection and Preparation

Chapter 28: Data Collection

- CSV and Excel sources
- SQL databases
- APIs
- Web scraping
- Sensors
- Logs
- Public datasets
- Cloud storage
- Structured data
- Semi-structured data
- Unstructured data
- Data labeling
- Sampling
- Data privacy
- Collection bias

Chapter 29: Exploratory Data Analysis

Topics

- Understanding dataset shape
- Data types
- Summary statistics
- Target-variable analysis
- Univariate analysis
- Bivariate analysis
- Multivariate analysis
- Correlation analysis
- Missing-value analysis
- Outlier analysis
- Class imbalance
- Feature-target relationships
- Data leakage detection
- Writing business insights

Chapter 30: Data Cleaning

- Removing duplicates

- Correcting data types
- Handling incorrect values
- Standardizing categories
- String cleaning
- Date cleaning
- Missing-value treatment
- Outlier treatment

Missing-value methods

- Row or column removal
- Mean imputation
- Median imputation
- Mode imputation
- Forward fill
- Backward fill
- KNN imputation
- Iterative imputation
- Missing indicators

Outlier methods

- IQR
- Z-score
- Percentile capping
- Winsorization
- Log transformation
- Isolation Forest

Chapter 31: Data Preprocessing

Feature scaling

- Min-max scaling
- Standardization
- Robust scaling
- MaxAbs scaling
- Vector normalization

Min-max scaling:

[x'=\frac{x-x_{min}}{x_{max}-x_{min}}]

Standardization:

[z=\frac{x-\mu}{\sigma}]

Categorical encoding

- Label encoding
- Ordinal encoding
- One-hot encoding
- Frequency encoding
- Binary encoding
- Target encoding
- Hash encoding

Data splitting

- Training set
- Validation set
- Test set
- Random splitting
- Stratified splitting
- Group splitting
- Time-based splitting

Advanced preprocessing

- Binarization
- Discretization
- Pipelines
- ColumnTransformer
- Preventing data leakage
- Handling imbalanced data
- Oversampling
- Undersampling
- SMOTE

Chapter 32: Feature Engineering and Selection

Feature engineering

- Domain-based features
- Date and time features
- Text features
- Interaction features
- Polynomial features
- Ratio features

- Aggregation features
- Log transformation
- Power transformation
- Binning
- Cyclical encoding

Feature selection

Filter methods

- Correlation
- Chi-square
- ANOVA
- Mutual information
- Variance threshold

Wrapper methods

- Forward selection
- Backward elimination
- Recursive Feature Elimination

Embedded methods

- Lasso
- Decision-tree importance
- Random-forest importance
- Gradient-boosting importance

Phase 8: Supervised Machine Learning

Chapter 33: Linear Regression

Topics

- Simple linear regression
- Multiple linear regression
- Regression line
- Coefficients and intercept
- Ordinary Least Squares
- Residuals
- Cost function

- Gradient descent

Equation:

[\hat y=\beta_0+\beta_1x_1+\cdots+\beta_nx_n]

Assumptions

- Linearity
- Independence
- Homoscedasticity
- Normal residuals
- Low multicollinearity

Evaluation

- MAE
- MSE
- RMSE
- (R^2)
- Adjusted (R^2)

Chapter 34: Polynomial and Regularized Regression

Polynomial regression

- Polynomial features
- Nonlinear relationships
- Degree selection
- Overfitting risk

Ridge regression

[J=\text{MSE}+\lambda\sum_{j=1}^{n}\beta_j^2]

Lasso regression

[J=\text{MSE}+\lambda\sum_{j=1}^{n}|\beta_j|]

Elastic Net

- Combination of L1 and L2 penalties
- Feature selection
- Multicollinearity handling

Chapter 35: Logistic Regression

Topics

- Binary classification
- Multiclass classification
- Sigmoid function
- Probability threshold
- Log odds
- Decision boundary
- Binary cross-entropy

Sigmoid:

[\sigma(z)=\frac{1}{1+e^{-z}}]

Classification metrics

- Confusion matrix
- Accuracy
- Precision
- Recall
- Specificity
- F1-score
- ROC curve
- AUC
- Precision-recall curve
- Log loss
- Threshold tuning

Chapter 36: K-Nearest Neighbors

- Instance-based learning
- Distance metrics
- Selecting (K)
- Majority voting
- Weighted KNN
- Scaling requirement
- Curse of dimensionality
- Classification and regression

Chapter 37: Naive Bayes

- Bayes' theorem
- Conditional independence assumption
- Gaussian Naive Bayes
- Multinomial Naive Bayes
- Bernoulli Naive Bayes
- Text classification
- Probability smoothing

Chapter 38: Decision Trees

- Root, internal and leaf nodes
- Splitting
- Entropy
- Information gain
- Gini impurity
- Tree depth
- Minimum samples
- Pruning
- Classification trees
- Regression trees
- Feature importance

Entropy:

[H(S)=-\sum_{i=1}^{c}p_i\log_2p_i]

Gini impurity:

[G=1-\sum_{i=1}^{c}p_i^2]

Chapter 39: Random Forest

- Ensemble learning
- Bootstrap sampling
- Bagging
- Random feature selection
- Majority voting
- Prediction averaging
- Out-of-bag score
- Feature importance
- Advantages and limitations

- Important hyperparameters

Chapter 40: Support Vector Machine

- Hyperplane
- Support vectors
- Margin
- Hard and soft margins
- Regularization parameter (C)
- Kernel trick
- Linear kernel
- Polynomial kernel
- RBF kernel
- Gamma
- SVC and SVR

Phase 9: Ensemble Learning and Boosting

Chapter 41: Ensemble Fundamentals

- Hard and soft voting
- Bagging
- Boosting
- Stacking
- Blending
- Bias reduction
- Variance reduction
- Base learners
- Meta learners

Chapter 42: AdaBoost

- Weak learners
- Sample weights
- Sequential training
- Misclassified observations
- Learning rate
- Number of estimators

Chapter 43: Gradient Boosting

- Residual learning
- Sequential trees

- Gradient-based optimization
- Learning rate
- Tree depth
- Subsampling
- Overfitting control

Chapter 44: XGBoost

- Regularized boosting
- Gradient and Hessian
- Missing-value handling
- Tree pruning
- Parallel processing
- Early stopping
- Feature importance
- Hyperparameter tuning

Chapter 45: LightGBM

- Leaf-wise growth
- Histogram-based learning
- Categorical features
- Large-data training
- Overfitting controls

Chapter 46: CatBoost

- Ordered boosting
- Native categorical handling
- Target leakage prevention
- Categorical feature combinations
- Important parameters

Phase 10: Model Evaluation and Optimization

Chapter 47: Cross-Validation

- Holdout validation
- K-Fold
- Stratified K-Fold
- Group K-Fold
- Leave-One-Out
- Repeated K-Fold

- Time Series Split
- Nested cross-validation

Chapter 48: Hyperparameter Tuning

- Manual search
- Grid Search
- Randomized Search
- Bayesian optimization
- Optuna
- Early stopping
- Learning curves
- Validation curves

Chapter 49: Model Interpretation

- Global vs local interpretation
- Feature importance
- Permutation importance
- Partial Dependence Plot
- ICE plot
- SHAP
- LIME
- Interpretable models
- Explaining predictions to business users

Phase 11: Unsupervised Learning

Chapter 50: K-Means Clustering

- Centroids
- Cluster assignment
- Centroid updates
- Inertia
- Elbow method
- Silhouette score
- K-Means++
- Scaling
- Limitations

Objective:

[J=\sum_{i=1}^{K}\sum_{x\in C_i}|x-\mu_i|^2]

Chapter 51: Hierarchical Clustering

- Agglomerative clustering
- Divisive clustering
- Dendrogram
- Single linkage
- Complete linkage
- Average linkage
- Ward linkage

Chapter 52: DBSCAN

- Density-based clustering
- Epsilon
- Minimum samples
- Core points
- Border points
- Noise points
- Irregular clusters
- Selecting parameters

Chapter 53: Gaussian Mixture Models

- Probabilistic clustering
- Gaussian components
- Soft clustering
- Expectation-Maximization
- AIC and BIC

Chapter 54: Dimensionality Reduction

PCA

- Variance
- Covariance matrix
- Standardization
- Eigenvalues
- Eigenvectors
- Principal components
- Explained variance
- Reconstruction

Other methods

- Truncated SVD

- t-SNE
- UMAP
- Linear Discriminant Analysis
- Autoencoders

Phase 12: Time-Series Analysis

Chapter 55: Time-Series Fundamentals

- Time index
- Trend
- Seasonality
- Cycles
- Noise
- Stationarity
- Autocorrelation
- Partial autocorrelation
- Lag features
- Rolling statistics
- Differencing

Chapter 56: Forecasting Models

- Moving average
- Exponential smoothing
- Holt’s method
- Holt-Winters
- AR
- MA
- ARMA
- ARIMA
- SARIMA
- Prophet
- Regression-based forecasting
- Tree-based forecasting
- LSTM forecasting

Evaluation

- MAE
- RMSE
- MAPE
- sMAPE

- Walk-forward validation

Phase 13: Recommendation Systems

Chapter 57: Recommendation Fundamentals

- Popularity-based recommendation
- Content-based filtering
- Collaborative filtering
- User-user similarity
- Item-item similarity
- Explicit and implicit feedback
- Cold-start problem
- Sparsity problem

Chapter 58: Advanced Recommendation

- Matrix factorization
- SVD
- Alternating Least Squares
- Neural collaborative filtering
- Hybrid recommendation
- Candidate generation
- Ranking
- Precision@K
- Recall@K
- MAP@K
- NDCG

Phase 14: Deep Learning

Chapter 59: Neural Network Foundations

- Biological vs artificial neurons
- Inputs, weights and bias
- Weighted sum
- Activation functions
- Layers
- Forward propagation
- Loss calculation
- Backpropagation
- Gradient descent

- Epoch
- Batch
- Iteration
- Learning rate

Neuron:

[z=\sum_{i=1}^{n}w_ix_i+b]

[a=f(z)]

Chapter 60: Activation and Loss Functions

Activation functions

- Linear
- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- PReLU
- ELU
- GELU
- Softmax

Loss functions

- Mean Squared Error
- Mean Absolute Error
- Binary cross-entropy
- Categorical cross-entropy
- Sparse categorical cross-entropy
- Hinge loss
- Huber loss

Chapter 61: Training Neural Networks

- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent
- Momentum
- RMSProp
- Adam
- AdamW

- Weight initialization
- Xavier initialization
- He initialization
- Vanishing gradients
- Exploding gradients
- Batch normalization
- Layer normalization
- Dropout
- L1 and L2 regularization
- Early stopping
- Learning-rate scheduling
- Gradient clipping

Chapter 62: TensorFlow, Keras and PyTorch

TensorFlow/Keras

- Tensors
- Sequential API
- Functional API
- Custom layers
- Model compilation
- Training and validation
- Callbacks
- Saving and loading models
- TensorBoard
- Custom training loops

PyTorch

- Tensors
- Datasets
- DataLoaders
- nn.Module
- Forward pass
- Autograd
- Loss functions
- Optimizers
- Training loops
- GPU training
- Model serialization

Phase 15: Computer Vision

Chapter 63: Image Processing

- Pixels and channels
- RGB and grayscale
- Image dimensions
- Resizing
- Cropping
- Rotation
- Thresholding
- Blurring
- Edge detection
- Contours
- Morphological operations
- Histograms
- OpenCV fundamentals
- Image augmentation

Chapter 64: Convolutional Neural Networks

- Convolution
- Kernels and filters
- Feature maps
- Stride
- Padding
- Pooling
- Receptive field
- Flattening
- Fully connected layers
- CNN training

Output size:

$$ \text{Output}=\left\lfloor\frac{N+2P-K}{S}\right\rfloor+1 $$

Where:

- (N): input size
- (P): padding
- (K): kernel size

- (S): stride

Chapter 65: Advanced Computer Vision

- Transfer learning
- Fine-tuning
- VGG
- ResNet
- Inception
- EfficientNet
- MobileNet
- Image classification
- Object detection
- R-CNN
- Fast R-CNN
- Faster R-CNN
- SSD
- YOLO
- Semantic segmentation
- Instance segmentation
- U-Net
- Mask R-CNN
- Vision Transformers
- Image captioning

Phase 16: Natural Language Processing

Chapter 66: NLP Fundamentals

- Text cleaning
- Lowercasing
- Tokenization
- Sentence segmentation
- Stop-word removal
- Stemming
- Lemmatization
- Part-of-speech tagging
- Named-entity recognition
- Dependency parsing
- N-grams
- Bag of Words
- TF-IDF

- Text classification
- Sentiment analysis

TF-IDF:

[TFIDF(t,d)=TF(t,d)\times \log\left(\frac{N}{DF(t)}\right)]

Chapter 67: Word Embeddings

- One-hot vectors
- Dense vectors
- Distributional semantics
- Word2Vec
- CBOW
- Skip-gram
- GloVe
- FastText
- Document embeddings
- Sentence embeddings
- Cosine similarity
- Contextual embeddings

Chapter 68: Sequence Models

- Sequential data
- Recurrent Neural Networks
- Hidden states
- Backpropagation Through Time
- Vanishing-gradient problem
- LSTM
- Input, forget and output gates
- GRU
- Bidirectional RNN
- Encoder-decoder architecture
- Attention mechanism
- Beam search

Phase 17: Transformers and Large Language Models

Chapter 69: Transformer Architecture

- Limitations of RNNs
- Self-attention

- Queries, keys and values
- Scaled dot-product attention
- Multi-head attention
- Positional encoding
- Feed-forward networks
- Residual connections
- Layer normalization
- Encoder
- Decoder
- Attention masks

Attention:

$$\text{Attention}(Q,K,V) =\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Chapter 70: Pretrained Language Models

- Language modeling
- Masked language modeling
- Causal language modeling
- BERT
- RoBERTa
- GPT
- T5
- Encoder-only models
- Decoder-only models
- Encoder-decoder models
- Tokenizers
- Subword tokenization
- Byte Pair Encoding
- WordPiece
- SentencePiece

Chapter 71: LLM Training and Adaptation

- Pretraining
- Supervised fine-tuning
- Instruction tuning
- Preference optimization
- RLHF
- DPO
- Prompt tuning
- Prefix tuning
- Adapters

- LoRA
- QLoRA
- Quantization
- Distillation
- Catastrophic forgetting
- Model alignment

Phase 18: Prompt Engineering and Generative AI

Chapter 72: Prompt Engineering

- System, user and assistant messages
- Zero-shot prompting
- One-shot prompting
- Few-shot prompting
- Role prompting
- Contextual prompting
- Structured-output prompting
- Chain-of-thought concepts
- Decomposition
- Self-consistency
- ReAct
- Tool-use prompting
- Prompt templates
- Prompt injection
- Prompt evaluation

Chapter 73: Generative AI

- Generative vs discriminative models
- Autoregressive generation
- Temperature
- Top-k sampling
- Top-p sampling
- Maximum tokens
- Frequency penalties
- Hallucination
- Grounding
- Multimodal models
- Text generation
- Image generation
- Audio generation

- Video generation

Chapter 74: Generative Image Models

- Autoencoders
- Variational Autoencoders
- GANs
- Generator and discriminator
- GAN training instability
- Conditional GAN
- Diffusion models
- Forward diffusion
- Reverse diffusion
- Noise prediction
- Latent diffusion
- Text conditioning

Phase 19: Embeddings, Vector Databases and RAG

Chapter 75: Embeddings

- Text embeddings
- Image embeddings
- Multimodal embeddings
- Embedding dimensions
- Semantic similarity
- Normalization
- Cosine similarity
- Dot-product similarity
- Euclidean distance
- Embedding-model selection

Chapter 76: Vector Databases

- Vector indexes
- Exact nearest-neighbor search
- Approximate nearest-neighbor search
- HNSW
- IVF
- Metadata
- Metadata filtering
- Collections and namespaces
- Chroma

- FAISS
- Pinecone
- Weaviate
- Qdrant
- Milvus

Chapter 77: Retrieval-Augmented Generation

Workflow

```
Documents → Chunks → Embeddings → Vector database
                                        ↓
Question → Query embedding → Retrieval
                                        ↓
                                Relevant context
                                        ↓
                                LLM → Answer
```

Topics

- Document loaders
- PDF, Word, web and database ingestion
- Text cleaning
- Chunk size
- Chunk overlap
- Recursive splitting
- Semantic chunking
- Embedding creation
- Indexing
- Query embeddings
- Top-K retrieval
- Similarity thresholds
- Metadata filtering
- Prompt construction
- Context-window management
- Source citations
- Conversation memory

Chapter 78: Advanced RAG

- Query rewriting
- Query expansion
- Multi-query retrieval

- Hybrid retrieval
- BM25
- Dense and sparse retrieval
- Reciprocal Rank Fusion
- Reranking
- Cross-encoders
- Parent-document retrieval
- Contextual compression
- Multi-vector retrieval
- Knowledge-graph RAG
- Corrective RAG
- Self-RAG
- Agentic RAG
- Multimodal RAG
- SQL RAG
- Graph RAG
- RAG caching
- RAG security

Chapter 79: RAG Evaluation

Retrieval metrics

- Precision@K
- Recall@K
- Mean Reciprocal Rank
- NDCG
- Hit rate
- Context precision
- Context recall

Generation metrics

- Answer relevance
- Faithfulness
- Groundedness
- Correctness
- Completeness
- Hallucination rate
- Citation accuracy
- Human evaluation
- LLM-as-a-judge

- RAGAS

Phase 20: AI Agents

Chapter 80: AI Agent Fundamentals

- Agent, environment and goal
- Observation
- Reasoning
- Planning
- Action
- Tool calling
- Memory
- Reflection
- Agent loop
- Deterministic vs autonomous workflows

Chapter 81: Advanced Agents

- ReAct agents
- Plan-and-execute
- Router agents
- Supervisor agents
- Multi-agent systems
- Short-term memory
- Long-term memory
- Episodic memory
- Semantic memory
- Human-in-the-loop
- Agent guardrails
- Retry and fallback logic
- Tool permissions
- Agent evaluation
- LangChain
- LangGraph
- LlamaIndex
- Model Context Protocol concepts

Phase 21: Reinforcement Learning

Chapter 82: Reinforcement Learning Fundamentals

- Agent
- Environment
- State
- Action
- Reward
- Policy
- Value function
- Episode
- Discount factor
- Markov Decision Process
- Exploration vs exploitation
- Return

Discounted return:

[G_t=\sum_{k=0}^{\infty}\gamma^kR_{t+k+1}]

Chapter 83: RL Algorithms

- Multi-armed bandits
- Epsilon-greedy
- Dynamic programming
- Monte Carlo methods
- Temporal-difference learning
- Q-learning
- SARSA
- Deep Q-Network
- Experience replay
- Target networks
- Policy gradients
- Actor-Critic
- A2C
- A3C
- PPO
- Reward shaping
- Offline reinforcement learning

Phase 22: ML Deployment

Chapter 84: Model Serialization and APIs

- Pickle
- Joblib
- Saving preprocessing pipelines
- Loading models
- REST APIs
- HTTP methods
- JSON
- Request and response
- Status codes
- FastAPI
- Pydantic
- Flask basics
- API validation
- Exception handling
- CORS
- API testing
- Swagger documentation

Chapter 85: ML User Interfaces

- Streamlit
- Gradio
- Input forms
- File uploads
- Prediction results
- Visualization
- Session state
- Model caching
- User validation
- Error handling

Chapter 86: Docker and Cloud Deployment

- Container concepts
- Images and containers
- Dockerfile
- Build and run
- Ports
- Volumes

- Environment variables
- Docker Compose
- Cloud basics
- Render
- AWS
- Azure
- Google Cloud
- Serverless deployment
- CPU vs GPU deployment
- Secrets management
- Logging

Phase 23: MLOps

Chapter 87: MLOps Fundamentals

- ML development vs production ML
- Reproducibility
- Data versioning
- Model versioning
- Experiment tracking
- Model registry
- CI/CD/CT
- Automated testing
- Feature stores
- Training pipelines
- Deployment pipelines

Tools

- Git and GitHub
- DVC
- MLflow
- Weights & Biases
- Airflow
- Prefect
- Kubeflow
- Docker
- Kubernetes

- GitHub Actions

Chapter 88: Monitoring and Maintenance

- Model performance monitoring
- System-performance monitoring
- Input-data validation
- Data drift
- Concept drift
- Prediction drift
- Feature drift
- Bias monitoring
- Latency
- Throughput
- Error rates
- Alerting
- Retraining triggers
- Champion-challenger testing
- Shadow deployment
- Canary deployment
- A/B testing
- Rollback strategies

Phase 24: Responsible AI and Security

Chapter 89: Responsible AI

- Fairness
- Bias
- Transparency
- Explainability
- Accountability
- Privacy
- Consent
- Data governance
- Human oversight
- Accessibility
- Environmental impact
- Model cards

- Datasheets for datasets

Chapter 90: AI Security

- Adversarial examples
- Data poisoning
- Model extraction
- Model inversion
- Membership inference
- Prompt injection
- Jailbreaking
- Indirect prompt injection
- Sensitive-data leakage
- Insecure tool use
- RAG poisoning
- Access control
- Input validation
- Output filtering
- Rate limiting
- Audit logging
- Secrets management

Phase 25: Software Engineering for ML

Chapter 91: Git and GitHub

- Repository
- Commit
- Branch
- Merge
- Pull request
- Clone
- Pull and push
- Merge conflicts
- .gitignore
- Semantic commits
- Branching strategies
- Code reviews
- GitHub Actions

Chapter 92: Clean ML Code

- Modular programming

- Separation of concerns
- Configuration files
- Environment variables
- Logging
- Type hints
- Docstrings
- Unit tests
- Integration tests
- Data tests
- Model tests
- Project packaging
- Dependency management
- Reproducible environments
- Design patterns for ML

Recommended structure

```
project/
├── data/
├── notebooks/
├── src/
│   ├── data/
│   ├── features/
│   ├── models/
│   └── evaluation/
├── tests/
├── app/
├── models/
├── config/
├── requirements.txt
├── Dockerfile
└── README.md
```

Phase 26: Specialized Advanced Topics

Chapter 93: Anomaly Detection

- Statistical anomaly detection
- Z-score
- IQR
- Isolation Forest
- Local Outlier Factor
- One-Class SVM

- Autoencoder-based detection
- Fraud detection
- Equipment-failure detection

Chapter 94: Graph Machine Learning

- Graphs, nodes and edges
- Adjacency matrix
- Degree
- Centrality
- Community detection
- Node embeddings
- Graph convolutional networks
- Graph attention networks
- Link prediction
- Knowledge graphs

Chapter 95: Causal Machine Learning

- Correlation vs causation
- Confounders
- Treatment and outcome
- Potential outcomes
- Average Treatment Effect
- Randomized experiments
- Propensity scores
- Difference-in-differences
- Instrumental variables
- Causal graphs
- Uplift modelling

Chapter 96: Multimodal AI

- Text-image models
- Vision-language models
- Image captioning
- Visual question answering
- Speech-to-text
- Text-to-speech
- Audio understanding
- Cross-modal embeddings
- Multimodal RAG

- Multimodal agents

Chapter 97: Edge AI and Model Optimization

- Model compression
- Pruning
- Quantization
- Knowledge distillation
- ONNX
- TensorRT
- TensorFlow Lite
- Mobile deployment
- Edge-device limitations
- Latency and memory optimization

Phase 27: AI/ML Projects

Beginner projects

1. Student marks predictor
2. House-price prediction
3. Salary predictor
4. Iris flower classification
5. Spam-message classifier
6. Customer segmentation
7. Sales-data analysis
8. Movie recommendation system

Intermediate projects

1. Customer churn prediction
2. Loan approval prediction
3. Credit-risk prediction
4. Fraud detection
5. Disease prediction
6. Employee attrition prediction
7. Retail-sales dashboard
8. Crop-yield prediction
9. Sentiment analysis

10. Time-series sales forecasting

Advanced projects

1. End-to-end MLOps prediction system
2. Real-time fraud-detection API
3. Object-detection application
4. Medical-image classifier
5. Multilingual NLP system
6. Recommendation engine
7. Document-question answering RAG
8. AI research assistant
9. SQL analytics agent
10. Multimodal RAG chatbot
11. Customer-support AI agent
12. Production model-monitoring system

Phase 28: Interview Preparation

Chapter 98: Python Interview Preparation

- Mutable vs immutable
- List vs tuple
- Set vs dictionary
- Shallow vs deep copy
- Iterators vs generators
- Decorators
- Closures
- OOP concepts
- Exception handling
- Memory management
- Multithreading vs multiprocessing
- GIL
- Coding problems

Chapter 99: Machine Learning Interview Preparation

- Bias-variance tradeoff
- Underfitting vs overfitting
- L1 vs L2
- Bagging vs boosting
- Classification vs regression
- Generative vs discriminative models

- Feature selection
- Cross-validation
- Data leakage
- Class imbalance
- Model selection
- Evaluation metrics
- Hyperparameter tuning
- Explainability
- Production monitoring

Chapter 100: Project and System-Design Interviews

Project explanation order

1. Problem statement
2. Business value
3. Dataset
4. Data cleaning
5. EDA findings
6. Feature engineering
7. Model selection
8. Evaluation metrics
9. Results
10. Deployment
11. Challenges
12. Future improvements

ML system-design topics

- Requirement clarification
- Data collection
- Label creation
- Feature pipeline
- Training pipeline
- Serving architecture
- Batch vs real-time inference
- Scalability
- Latency
- Monitoring
- Retraining
- Security
- Cost

Recommended Learning Order

Follow this order without skipping foundations:

1. Python fundamentals
2. Python data structures and OOP
3. NumPy and Pandas
4. Data visualization
5. SQL
6. Statistics and probability
7. Linear algebra and calculus
8. Data cleaning and EDA
9. Feature engineering
10. Classical machine learning
11. Model evaluation and tuning
12. Unsupervised learning
13. Time series and recommendations
14. Deep-learning fundamentals
15. Computer vision
16. NLP
17. Transformers and LLMs
18. RAG and vector databases
19. AI agents
20. Deployment
21. Docker and cloud
22. MLOps and monitoring
23. Responsible AI and security
24. End-to-end projects
25. Interview and system-design preparation

Practical Study Rule for Every Chapter

For each chapter, complete these seven activities:

1. Learn the definition and intuition.
2. Understand important formulas.
3. Write basic code without copying.
4. Perform a line-by-line dry run.
5. Build one small practical example.
6. Solve interview questions.

7. Add the topic to an end-to-end project.

This 100-chapter path covers the complete progression from a Python beginner to a job-ready AI/ML engineer.

You said:

AIML complete Chapter and each Topicwise syllabus

ChatGPT said:

Complete AI/ML Syllabus: Basic to Advanced

This is a chapter-wise and topic-wise roadmap for becoming a job-ready AI/ML Engineer. Study the chapters in the given order.

Phase 1: AI, ML and Data Science Foundations

Chapter 1: Introduction to Artificial Intelligence

- Meaning and definition of AI
- History and evolution of AI
- Goals of artificial intelligence
- Human intelligence vs machine intelligence
- AI vs ML vs Deep Learning vs Data Science
- Intelligent systems
- Intelligent agents
- Agent, environment, state, action and reward
- Rational agents
- PEAS representation
- Applications of AI
- Benefits and limitations of AI
- Ethical and responsible AI

Chapter 2: Types of Artificial Intelligence

Based on capability

- Artificial Narrow Intelligence
- Artificial General Intelligence
- Artificial Super Intelligence

Based on functionality

- Reactive machines

- Limited-memory AI
- Theory-of-mind AI
- Self-aware AI

Chapter 3: AI Problem-Solving Techniques

- Problem formulation
- State-space representation
- Initial and goal states
- Operators and actions
- Search trees and search graphs
- Breadth-First Search
- Depth-First Search
- Uniform-Cost Search
- Depth-Limited Search
- Iterative Deepening
- Bidirectional Search
- Greedy Best-First Search
- A* search
- Heuristic functions
- Hill climbing
- Beam search
- Constraint satisfaction problems
- Minimax algorithm
- Alpha-beta pruning

Chapter 4: Introduction to Machine Learning

- Definition of machine learning
- Traditional programming vs machine learning
- Data, samples and observations
- Independent and dependent variables
- Features and target
- Model and algorithm
- Training and prediction
- Parameters and hyperparameters
- Loss and cost functions
- Training, validation and test data
- Generalization
- Inference

- Model lifecycle

Chapter 5: Types of Machine Learning

- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Self-supervised learning
- Reinforcement learning
- Online learning
- Batch learning
- Active learning
- Transfer learning
- Federated learning
- Multi-task learning

Chapter 6: Common Machine-Learning Problems

- Regression
- Binary classification
- Multiclass classification
- Multilabel classification
- Clustering
- Dimensionality reduction
- Anomaly detection
- Recommendation
- Ranking
- Time-series forecasting
- Natural-language processing
- Computer vision

Chapter 7: Complete AI/ML Lifecycle

- Business-problem understanding
- Problem framing
- Defining inputs and outputs
- Selecting business metrics
- Data collection
- Data validation
- Exploratory Data Analysis
- Data cleaning
- Feature engineering
- Model selection
- Model training

- Model evaluation
- Hyperparameter tuning
- Model deployment
- Monitoring
- Retraining
- Documentation

Phase 2: Python Programming

Chapter 8: Python Environment and Setup

- Python installation
- Python interpreter
- Running Python scripts
- Jupyter Notebook
- Google Colab
- VS Code
- Anaconda and Conda
- pip
- Virtual environments
- Python packages
- requirements.txt

Chapter 9: Python Syntax and Fundamentals

- Python syntax
- Indentation
- Comments
- Keywords
- Identifiers
- Statements
- Expressions
- Variables
- Constants
- Naming conventions
- Dynamic typing
- Multiple assignment
- type()

- id()

Chapter 10: Python Data Types

Numeric types

- Integer
- Float
- Complex
- Boolean

Collection and other types

- String
- List
- Tuple
- Set
- Frozen set
- Dictionary
- Range
- Bytes
- Byte array
- None

Additional concepts

- Mutable and immutable objects
- Hashable and unhashable objects
- Type checking
- Type conversion
- Implicit conversion
- Explicit conversion

Chapter 11: Python Operators

- Arithmetic operators
- Comparison operators
- Assignment operators
- Logical operators
- Identity operators
- Membership operators
- Bitwise operators
- Walrus operator
- Operator precedence

- Associativity

Chapter 12: Input and Output

- input()
- print()
- sep
- end
- Escape sequences
- String formatting
- f-strings
- format()
- Percentage formatting
- Formatting numbers
- Reading multiple inputs
- Input validation

Chapter 13: Conditional Statements

- if
- if-else
- if-elif-else
- Nested conditions
- Multiple conditions
- Logical conditions
- Ternary expressions
- match-case
- Truthy and falsy values

Chapter 14: Loops

- for loop
- while loop
- Nested loops
- Infinite loops
- range()
- break
- continue
- pass
- Loop else
- enumerate()
- zip()

- Pattern programs

Chapter 15: Python Strings

- String creation
- Indexing
- Negative indexing
- Slicing
- String immutability
- String traversal
- Concatenation
- Repetition
- Membership
- String comparison
- Unicode
- Raw strings

String methods

- upper(), lower()
- title(), capitalize()
- strip(), lstrip(), rstrip()
- split(), rsplit(), join()
- find(), rfind(), index()
- replace(), count()
- startswith(), endswith()
- isalpha(), isdigit(), isalnum()
- partition()
- String validation

String problems

- Reverse a string
- Palindrome
- Anagram
- Character frequency
- Remove duplicates
- First non-repeating character
- String compression
- Longest substring

Chapter 16: Python Lists

- List creation

- Indexing and slicing
- Updating elements
- Nested lists
- List traversal
- List comparison
- List aliasing
- Shallow copying
- Deep copying
- List comprehensions
- Nested comprehensions

List methods

- append()
- extend()
- insert()
- remove()
- pop()
- clear()
- index()
- count()
- sort()
- reverse()
- copy()

Chapter 17: Tuples

- Tuple creation
- Single-element tuple
- Indexing and slicing
- Tuple immutability
- Packing
- Unpacking
- Extended unpacking
- Nested tuples
- Tuple methods
- Returning multiple values
- List vs tuple
- Named tuples

Chapter 18: Sets

- Set creation
- Unique values

- Adding and deleting values
- Union
- Intersection
- Difference
- Symmetric difference
- Subset
- Superset
- Disjoint sets
- Set comprehension
- Frozen sets
- Set vs list

Chapter 19: Dictionaries

- Key-value pairs
- Creating dictionaries
- Accessing and updating values
- Adding and deleting items
- Dictionary traversal
- Nested dictionaries
- Dictionary comprehensions
- Hashable keys
- Dictionary merging

Dictionary methods

- get()
- keys()
- values()
- items()
- update()
- pop()
- popitem()
- setdefault()
- fromkeys()

Chapter 20: Python Functions

- Function definition
- Function calling
- Parameters and arguments
- Positional arguments
- Keyword arguments
- Default arguments

- Positional-only parameters
- Keyword-only parameters
- *args
- **kwargs
- Return statement
- Multiple return values
- Docstrings
- Type annotations

Chapter 21: Scope and Functional Programming

- Local scope
- Global scope
- Enclosing scope
- Built-in scope
- LEGB rule
- global
- nonlocal
- Lambda functions
- First-class functions
- Higher-order functions
- map()
- filter()
- reduce()
- any()
- all()
- Closures
- Decorators
- Recursion

Chapter 22: Object-Oriented Programming

- Class and object
- Attributes and methods
- Constructor
- self
- Instance variables
- Class variables
- Instance methods
- Class methods

- Static methods

Four pillars of OOP

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Advanced OOP

- Single inheritance
- Multiple inheritance
- Multilevel inheritance
- Hierarchical inheritance
- Hybrid inheritance
- Method overriding
- Operator overloading
- Abstract classes
- super()
- Method Resolution Order
- Composition
- Aggregation
- Properties
- Getters and setters
- Magic methods
- Dataclasses

Chapter 23: Exception Handling

- Syntax errors
- Runtime errors
- Logical errors
- try
- except
- Multiple exceptions
- else
- finally
- raise
- Assertions
- Custom exceptions
- Exception propagation

- Exception hierarchy

Chapter 24: File Handling

- Text files
- Binary files
- File modes
- Opening and closing files
- read()
- readline()
- readlines()
- write()
- writelines()
- File pointer
- seek()
- tell()
- with statement
- CSV files
- JSON files
- Pickle files
- Path handling

Chapter 25: Modules and Packages

- Importing modules
- import
- from...import
- Aliases
- User-defined modules
- Built-in modules
- Creating packages
- __init__.py
- __name__
- __main__
- Relative and absolute imports
- Package installation

Chapter 26: Advanced Python

- Iterators
- Iterables
- Generators
- Generator expressions
- Decorators

- Closures
- Context managers
- Regular expressions
- Type hints
- Logging
- Unit testing
- Memory management
- Garbage collection
- Shallow and deep copying
- Multithreading
- Multiprocessing
- Asynchronous programming
- async and await
- API requests
- Web scraping

Phase 3: Data Structures and Algorithms

Chapter 27: Complexity Analysis

- Algorithm analysis
- Time complexity
- Space complexity
- Big O
- Big Omega
- Big Theta
- Best, average and worst cases
- Constant time
- Logarithmic time
- Linear time
- Linearithmic time
- Quadratic time
- Exponential time
- Recursive complexity
- Amortized analysis

Chapter 28: Arrays and Problem Patterns

- Array traversal
- Searching
- Sorting
- Prefix sum

- Two pointers
- Sliding window
- Hashing
- Kadane’s algorithm
- Binary search
- Matrix traversal

Chapter 29: Linked Lists, Stacks and Queues

- Singly linked list
- Doubly linked list
- Circular linked list
- Stack operations
- Queue operations
- Circular queue
- Deque
- Priority queue
- Monotonic stack
- Applications

Chapter 30: Trees, Heaps and Graphs

- Binary trees
- Binary Search Trees
- Tree traversal
- AVL-tree concepts
- Heaps
- Tries
- Graph representation
- BFS
- DFS
- Topological sorting
- Shortest paths
- Minimum spanning trees

Chapter 31: Algorithmic Techniques

- Recursion
- Backtracking
- Greedy algorithms
- Divide and conquer
- Dynamic programming
- Memoization
- Tabulation

- Graph algorithms
- String algorithms

Phase 4: Mathematics for AI/ML

Chapter 32: Basic Mathematics

- Functions
- Linear functions
- Nonlinear functions
- Exponents
- Logarithms
- Summation notation
- Sequences
- Series
- Polynomials
- Coordinate geometry
- Distance formulas

Chapter 33: Linear Algebra Fundamentals

- Scalars
- Vectors
- Matrices
- Tensors
- Shapes and dimensions
- Vector operations
- Dot product
- Cross product
- Matrix operations
- Matrix multiplication
- Transpose
- Identity matrix
- Diagonal matrix
- Symmetric matrix

Chapter 34: Advanced Linear Algebra

- Matrix inverse
- Determinant
- Rank
- Linear independence
- Basis

- Vector spaces
- Orthogonality
- Norms
- Eigenvalues
- Eigenvectors
- Eigendecomposition
- Singular Value Decomposition
- Matrix factorization
- Cosine similarity

Chapter 35: Calculus

- Limits
- Continuity
- Derivatives
- Partial derivatives
- Product rule
- Quotient rule
- Chain rule
- Gradients
- Directional derivatives
- Jacobian matrix
- Hessian matrix
- Maxima and minima
- Convex functions
- Integration
- Gradient descent
- Backpropagation mathematics

Chapter 36: Probability Fundamentals

- Random experiments
- Sample spaces
- Events
- Probability rules
- Independent events
- Dependent events
- Mutually exclusive events
- Joint probability
- Marginal probability
- Conditional probability
- Bayes' theorem

- Random variables

Chapter 37: Probability Distributions

- Probability Mass Function
- Probability Density Function
- Cumulative Distribution Function
- Bernoulli distribution
- Binomial distribution
- Uniform distribution
- Normal distribution
- Poisson distribution
- Exponential distribution
- Multinomial distribution
- Gaussian mixture distribution
- Expectation
- Variance

Chapter 38: Descriptive Statistics

- Population and sample
- Mean
- Median
- Mode
- Weighted mean
- Range
- Variance
- Standard deviation
- Quartiles
- Percentiles
- Interquartile range
- Skewness
- Kurtosis
- Covariance
- Correlation

Chapter 39: Inferential Statistics

- Sampling methods
- Sampling distributions
- Central Limit Theorem
- Standard error
- Point estimation
- Interval estimation

- Confidence intervals
- Hypothesis testing
- Null and alternative hypotheses
- Significance level
- p-value
- Type I and Type II errors
- Statistical power

Chapter 40: Statistical Tests

- Z-test
- One-sample t-test
- Independent t-test
- Paired t-test
- Chi-square test
- One-way ANOVA
- Two-way ANOVA
- Mann–Whitney U test
- Wilcoxon test
- Kruskal–Wallis test
- A/B testing

Phase 5: Data-Science Libraries

Chapter 41: NumPy Fundamentals

- NumPy arrays
- Dimensions
- Shapes
- Data types
- Array creation
- Indexing
- Slicing
- Boolean indexing
- Fancy indexing
- Copy vs view
- Reshaping
- Flattening
- Concatenation

- Splitting

Chapter 42: Advanced NumPy

- Broadcasting
- Vectorization
- Universal functions
- Aggregation
- Sorting
- Random module
- Missing values
- Linear algebra
- Determinants
- Matrix inverses
- Eigenvalues and eigenvectors
- SVD
- einsum
- Memory optimization

Chapter 43: Pandas Fundamentals

- Series
- DataFrame
- Index
- Creating DataFrames
- Reading CSV
- Reading Excel
- Reading JSON
- Reading SQL
- Data inspection
- Row and column selection
- loc
- iloc
- Boolean filtering
- query()

Chapter 44: Data Cleaning with Pandas

- Missing values
- Duplicate values
- Incorrect data types
- String cleaning
- Datetime processing
- Category standardization

- Outlier identification
- Replacing values
- Mapping
- Applying functions

Chapter 45: Pandas Aggregation and Combination

- Sorting
- GroupBy
- Aggregation
- Pivot tables
- Crosstabs
- Rolling operations
- Expanding operations
- Merge
- Join
- Concatenate
- MultiIndex
- Method chaining
- Memory optimization

Chapter 46: Matplotlib

- Figure and axes
- Line plots
- Bar plots
- Scatter plots
- Histograms
- Pie charts
- Box plots
- Area charts
- Stack plots
- Error bars
- Subplots
- Titles and labels
- Legends
- Annotations
- Styles
- Saving graphs

Chapter 47: Seaborn and Interactive Visualization

- Distribution plots
- Count plots

- Bar plots
- Box plots
- Violin plots
- Strip plots
- Swarm plots
- Regression plots
- Pair plots
- Joint plots
- Heatmaps
- FacetGrid
- Plotly
- Bokeh
- Dashboard design
- Data storytelling

Chapter 48: SciPy and Statsmodels

- Scientific constants
- Optimization
- Integration
- Interpolation
- Signal processing
- Distance calculations
- Statistical distributions
- Statistical tests
- Ordinary Least Squares
- Generalized linear models
- Time-series analysis
- Diagnostic tests

Phase 6: SQL and Databases

Chapter 49: Database Fundamentals

- Data and databases
- DBMS and RDBMS
- Tables, rows and columns
- Primary keys
- Foreign keys
- Candidate keys
- Composite keys
- Constraints

- Relationships
- Schema
- Normalization
- Denormalization
- ACID properties
- Transactions

Chapter 50: SQL Basics

- CREATE
- ALTER
- DROP
- TRUNCATE
- INSERT
- UPDATE
- DELETE
- SELECT
- WHERE
- ORDER BY
- DISTINCT
- LIMIT
- BETWEEN
- IN
- LIKE
- CASE

Chapter 51: SQL Aggregation and Joins

- Aggregate functions
- GROUP BY
- HAVING
- Inner join
- Left join
- Right join
- Full join
- Cross join
- Self join
- Multiple-table joins

Chapter 52: Advanced SQL

- Subqueries
- Correlated subqueries
- CTEs

- Recursive CTEs
- Views
- Indexes
- Stored procedures
- Functions
- Triggers
- Transactions
- Query optimization

Chapter 53: SQL Window Functions

- OVER
- PARTITION BY
- ROW_NUMBER
- RANK
- DENSE_RANK
- NTILE
- LAG
- LEAD
- FIRST_VALUE
- LAST_VALUE
- Running totals
- Moving averages

Chapter 54: NoSQL Databases

- SQL vs NoSQL
- Document databases
- Key-value databases
- Column databases
- Graph databases
- MongoDB
- Collections and documents
- MongoDB CRUD
- Aggregation pipeline
- MongoDB indexing

Phase 7: Data Collection, EDA and Preprocessing

Chapter 55: Data Collection

- CSV and Excel
- SQL databases

- REST APIs
- Web scraping
- Sensors and logs
- Public datasets
- Cloud storage
- Structured data
- Semi-structured data
- Unstructured data
- Data labeling
- Sampling
- Privacy
- Collection bias

Chapter 56: Exploratory Data Analysis

- Dataset shape
- Column understanding
- Data types
- Summary statistics
- Target analysis
- Univariate analysis
- Bivariate analysis
- Multivariate analysis
- Missing-value analysis
- Outlier analysis
- Correlation analysis
- Class imbalance
- Data leakage
- Business insights

Chapter 57: Missing Values and Duplicates

- Missing completely at random
- Missing at random
- Missing not at random
- Row deletion
- Column deletion
- Mean imputation
- Median imputation
- Mode imputation
- Forward and backward fill
- KNN imputation
- Iterative imputation

- Missing indicators
- Duplicate detection

Chapter 58: Outlier Detection and Treatment

- Box plots
- IQR method
- Z-score
- Modified Z-score
- Percentile method
- Winsorization
- Capping
- Transformation
- Isolation Forest
- Local Outlier Factor
- Domain-based treatment

Chapter 59: Feature Scaling and Transformation

- Min-max scaling
- Standardization
- Robust scaling
- MaxAbs scaling
- Vector normalization
- Log transformation
- Square-root transformation
- Box-Cox transformation
- Yeo-Johnson transformation
- Quantile transformation

Chapter 60: Categorical Encoding

- Label encoding
- Ordinal encoding
- One-hot encoding
- Dummy encoding
- Frequency encoding
- Count encoding
- Binary encoding
- Hash encoding
- Target encoding
- Leave-one-out encoding

- High-cardinality features

Chapter 61: Dataset Splitting and Leakage

- Training set
- Validation set
- Test set
- Random split
- Stratified split
- Group split
- Time-based split
- Data leakage
- Target leakage
- Train-test contamination
- Preprocessing leakage

Chapter 62: Imbalanced Data

- Class distribution
- Accuracy paradox
- Random oversampling
- Random undersampling
- SMOTE
- Borderline SMOTE
- ADASYN
- Tomek links
- SMOTEENN
- Class weights
- Threshold tuning
- Suitable evaluation metrics

Chapter 63: Feature Engineering

- Domain features
- Interaction features
- Polynomial features
- Ratio features
- Aggregate features
- Date and time features
- Text features
- Binning
- Cyclical encoding
- Lag features
- Rolling features

- Feature crosses

Chapter 64: Feature Selection

Filter methods

- Variance threshold
- Correlation
- Chi-square
- ANOVA
- Mutual information

Wrapper methods

- Forward selection
- Backward elimination
- Recursive Feature Elimination

Embedded methods

- Lasso
- Decision-tree importance
- Random-forest importance
- Boosting importance

Chapter 65: Scikit-learn Pipelines

- Estimators
- Transformers
- fit()
- transform()
- fit_transform()
- predict()
- Pipeline
- ColumnTransformer
- Feature unions
- Custom transformers
- Preventing leakage
- Saving complete pipelines

Phase 8: Machine-Learning Fundamentals

Chapter 66: Core ML Concepts

- Model
- Algorithm
- Parameters
- Hyperparameters
- Features
- Labels
- Objective function
- Loss function
- Cost function
- Optimizer
- Convergence
- Generalization

Chapter 67: Bias, Variance and Model Complexity

- Bias
- Variance
- Irreducible error
- Underfitting
- Overfitting
- Bias-variance trade-off
- Learning curves
- Regularization
- Model complexity
- Generalization error

Chapter 68: Gradient Descent

- Cost surface
- Gradient
- Learning rate
- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent
- Local and global minima
- Convergence
- Learning-rate scheduling
- Momentum

Phase 9: Regression Algorithms

Chapter 69: Linear Regression

- Simple linear regression
- Multiple linear regression
- Regression coefficients
- Intercept
- Ordinary Least Squares
- Residuals
- Cost function
- Gradient descent
- Predictions

Assumptions

- Linearity
- Independence
- Homoscedasticity
- Normal residuals
- Low multicollinearity

Chapter 70: Polynomial Regression

- Nonlinear relationships
- Polynomial features
- Interaction terms
- Degree selection
- Model complexity
- Overfitting control

Chapter 71: Regularized Regression

- Regularization
- L1 penalty
- L2 penalty
- Ridge regression
- Lasso regression
- Elastic Net
- Regularization strength
- Feature selection

- Multicollinearity

Chapter 72: Regression Evaluation

- Residual analysis
- MAE
- MSE
- RMSE
- RMSLE
- MAPE
- (R^2)
- Adjusted (R^2)
- Cross-validation
- Error interpretation

Phase 10: Classification Algorithms

Chapter 73: Logistic Regression

- Binary classification
- Multiclass classification
- Sigmoid function
- Log odds
- Probability threshold
- Decision boundary
- Binary cross-entropy
- Regularization
- One-vs-rest
- Multinomial logistic regression

Chapter 74: Classification Evaluation

- Confusion matrix
- True positive
- True negative
- False positive
- False negative
- Accuracy
- Precision
- Recall
- Specificity
- F1-score
- F-beta score

- ROC curve
- AUC
- Precision-recall curve
- Log loss
- Threshold tuning
- Macro, micro and weighted averaging

Chapter 75: K-Nearest Neighbors

- Instance-based learning
- Distance metrics
- Selecting (K)
- Majority voting
- Weighted KNN
- KNN regression
- Feature scaling
- Curse of dimensionality
- Advantages and limitations

Chapter 76: Naive Bayes

- Bayes' theorem
- Conditional independence
- Gaussian Naive Bayes
- Multinomial Naive Bayes
- Bernoulli Naive Bayes
- Complement Naive Bayes
- Probability smoothing
- Text classification

Chapter 77: Decision Trees

- Root, branch and leaf nodes
- Splitting
- Entropy
- Information gain
- Gini impurity
- Tree depth
- Minimum samples
- Maximum features
- Pruning
- Classification trees
- Regression trees

- Feature importance

Chapter 78: Support Vector Machines

- Hyperplanes
- Support vectors
- Margin
- Hard margin
- Soft margin
- Parameter (C)
- Kernel trick
- Linear kernel
- Polynomial kernel
- RBF kernel
- Gamma
- SVC
- SVR

Phase 11: Ensemble Learning

Chapter 79: Ensemble Fundamentals

- Ensemble learning
- Base models
- Homogeneous ensembles
- Heterogeneous ensembles
- Hard voting
- Soft voting
- Bagging
- Boosting
- Stacking
- Blending
- Bias and variance reduction

Chapter 80: Random Forest

- Bootstrap sampling
- Bagging
- Random feature selection
- Majority voting
- Prediction averaging
- Out-of-bag score
- Feature importance

- Classification and regression
- Hyperparameters

Chapter 81: AdaBoost

- Weak learners
- Sequential learning
- Sample weights
- Misclassified samples
- Estimator weights
- Learning rate
- Number of estimators

Chapter 82: Gradient Boosting

- Residual learning
- Sequential trees
- Gradients
- Learning rate
- Tree depth
- Number of trees
- Subsampling
- Regularization

Chapter 83: XGBoost

- Gradient and Hessian
- Regularized objective
- Missing-value handling
- Tree pruning
- Parallel processing
- Early stopping
- Feature importance
- Important hyperparameters

Chapter 84: LightGBM and CatBoost

LightGBM

- Leaf-wise growth
- Histogram learning
- Gradient-based sampling
- Exclusive feature bundling

- Categorical features

CatBoost

- Ordered boosting
- Native categorical handling
- Target-statistic encoding
- Leakage prevention
- Important parameters

Phase 12: Model Validation and Optimization

Chapter 85: Cross-Validation

- Holdout validation
- K-Fold
- Stratified K-Fold
- Repeated K-Fold
- Group K-Fold
- Leave-One-Out
- Time Series Split
- Nested cross-validation

Chapter 86: Hyperparameter Tuning

- Manual search
- Grid Search
- Randomized Search
- Bayesian optimization
- Optuna
- Early stopping
- Validation curves
- Learning curves
- Search-space design

Chapter 87: Model Interpretation

- Global interpretation
- Local interpretation
- Coefficient interpretation
- Tree feature importance
- Permutation importance
- Partial Dependence Plots

- ICE plots
- LIME
- SHAP
- Explaining predictions

Chapter 88: Probability Calibration

- Predicted probability
- Well-calibrated models
- Calibration curves
- Brier score
- Platt scaling
- Isotonic regression
- Decision thresholds
- Cost-sensitive decisions

Phase 13: Unsupervised Learning

Chapter 89: K-Means Clustering

- Centroids
- Cluster assignment
- Centroid update
- Inertia
- K-Means++
- Elbow method
- Silhouette score
- Cluster interpretation
- Scaling requirement
- Limitations

Chapter 90: Hierarchical Clustering

- Agglomerative clustering
- Divisive clustering
- Dendrogram
- Distance matrix
- Single linkage
- Complete linkage
- Average linkage

- Ward linkage

Chapter 91: DBSCAN and Density Clustering

- Density-based clustering
- Epsilon
- Minimum samples
- Core points
- Border points
- Noise points
- Arbitrary-shaped clusters
- OPTICS
- HDBSCAN concepts

Chapter 92: Gaussian Mixture Models

- Probabilistic clustering
- Gaussian components
- Soft clustering
- Expectation-Maximization
- Mixture weights
- Covariance types
- AIC
- BIC

Chapter 93: Principal Component Analysis

- High-dimensional data
- Standardization
- Covariance matrix
- Eigenvalues
- Eigenvectors
- Principal components
- Explained variance
- Component selection
- Reconstruction
- Visualization

Chapter 94: Advanced Dimensionality Reduction

- Truncated SVD
- Linear Discriminant Analysis
- Kernel PCA
- t-SNE

- UMAP
- Autoencoders
- Manifold learning

Chapter 95: Anomaly Detection

- Point anomalies
- Contextual anomalies
- Collective anomalies
- Statistical methods
- Z-score
- IQR
- Isolation Forest
- Local Outlier Factor
- One-Class SVM
- Autoencoder detection

Phase 14: Time-Series Analysis

Chapter 96: Time-Series Fundamentals

- Time index
- Trend
- Seasonality
- Cycles
- Noise
- Stationarity
- White noise
- Autocorrelation
- Partial autocorrelation
- Lag features
- Rolling statistics
- Differencing
- Decomposition

Chapter 97: Statistical Forecasting

- Naive forecasting
- Moving averages
- Exponential smoothing
- Holt’s method
- Holt-Winters
- Autoregression

- Moving-average model
- ARMA
- ARIMA
- SARIMA
- ACF and PACF

Chapter 98: Advanced Forecasting

- Prophet
- Regression forecasting
- Tree-based forecasting
- XGBoost forecasting
- LSTM forecasting
- Multivariate time series
- Walk-forward validation
- Forecast intervals
- MAE, RMSE, MAPE and sMAPE

Phase 15: Recommendation Systems

Chapter 99: Recommendation Fundamentals

- Popularity-based recommendation
- Content-based filtering
- Collaborative filtering
- User-user similarity
- Item-item similarity
- Explicit feedback
- Implicit feedback
- Cold-start problem
- Sparsity

Chapter 100: Advanced Recommendation

- User-item matrix
- Matrix factorization
- SVD
- Alternating Least Squares
- Neural collaborative filtering
- Hybrid systems
- Candidate generation
- Ranking
- Precision@K

- Recall@K
- MAP@K
- NDCG

Phase 16: Deep Learning

Chapter 101: Artificial Neural Networks

- Biological neuron
- Artificial neuron
- Inputs
- Weights
- Bias
- Weighted sum
- Activation
- Layers
- Forward propagation
- Loss calculation
- Backpropagation
- Epochs
- Batches
- Iterations

Chapter 102: Activation Functions

- Linear
- Binary step
- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- PReLU
- ELU
- SELU
- GELU
- Softmax
- Activation selection

Chapter 103: Neural-Network Loss Functions

- Mean Squared Error
- Mean Absolute Error
- Huber loss

- Binary cross-entropy
- Categorical cross-entropy
- Sparse categorical cross-entropy
- Hinge loss
- Contrastive loss
- Triplet loss

Chapter 104: Neural-Network Optimization

- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent
- Momentum
- Nesterov momentum
- Adagrad
- RMSProp
- Adam
- AdamW
- Learning-rate scheduling

Chapter 105: Improving Neural Networks

- Weight initialization
- Xavier initialization
- He initialization
- Vanishing gradients
- Exploding gradients
- Batch normalization
- Layer normalization
- Dropout
- L1 and L2 regularization
- Early stopping
- Gradient clipping
- Data augmentation

Chapter 106: TensorFlow and Keras

- Tensors
- Variables
- Sequential API
- Functional API
- Model subclassing
- Layers
- Compilation

- Training
- Validation
- Callbacks
- Saving and loading
- TensorBoard
- Custom training loops

Chapter 107: PyTorch

- Tensors
- Tensor operations
- Autograd
- Datasets
- DataLoaders
- nn.Module
- Forward method
- Loss functions
- Optimizers
- Training loops
- Validation loops
- GPU training
- Saving models

Phase 17: Computer Vision

Chapter 108: Image Processing

- Pixels
- Image dimensions
- RGB
- Grayscale
- Color spaces
- Resizing
- Cropping
- Rotation
- Thresholding
- Blurring
- Edge detection
- Contours
- Histograms
- Morphological operations

- OpenCV

Chapter 109: Convolutional Neural Networks

- Convolution
- Kernels
- Filters
- Feature maps
- Stride
- Padding
- Pooling
- Receptive field
- Flattening
- Fully connected layers
- CNN architecture
- CNN training

Chapter 110: CNN Architectures and Transfer Learning

- LeNet
- AlexNet
- VGG
- Inception
- ResNet
- DenseNet
- MobileNet
- EfficientNet
- Transfer learning
- Feature extraction
- Fine-tuning

Chapter 111: Object Detection

- Bounding boxes
- Intersection over Union
- Non-Maximum Suppression
- Anchor boxes
- R-CNN
- Fast R-CNN
- Faster R-CNN
- SSD
- YOLO

- Mean Average Precision

Chapter 112: Image Segmentation

- Semantic segmentation
- Instance segmentation
- Panoptic segmentation
- Encoder-decoder networks
- Fully Convolutional Networks
- U-Net
- Mask R-CNN
- Segmentation metrics
- IoU
- Dice coefficient

Chapter 113: Advanced Computer Vision

- Vision Transformers
- Image captioning
- Facial recognition
- Optical Character Recognition
- Pose estimation
- Image generation
- Image similarity
- Visual question answering

Phase 18: Natural Language Processing

Chapter 114: NLP Fundamentals

- Corpus
- Document
- Sentence
- Token
- Vocabulary
- Text cleaning
- Lowercasing
- Tokenization
- Stop-word removal
- Stemming
- Lemmatization

- N-grams

Chapter 115: Linguistic Processing

- Part-of-speech tagging
- Named-entity recognition
- Dependency parsing
- Constituency parsing
- Chunking
- Coreference resolution
- Language detection
- Text normalization

Chapter 116: Text Representation

- One-hot encoding
- Bag of Words
- Count vectors
- TF-IDF
- Character features
- Word features
- N-gram features
- Sparse matrices
- Feature hashing

Chapter 117: Word and Sentence Embeddings

- Dense embeddings
- Word2Vec
- CBOW
- Skip-gram
- GloVe
- FastText
- Document embeddings
- Sentence embeddings
- Contextual embeddings
- Cosine similarity

Chapter 118: NLP Applications

- Text classification
- Sentiment analysis
- Spam detection
- Topic modelling

- Named-entity recognition
- Question answering
- Text summarization
- Machine translation
- Chatbots
- Information extraction

Chapter 119: Topic Modelling

- Latent topics
- Document-term matrix
- Latent Semantic Analysis
- Latent Dirichlet Allocation
- Non-negative Matrix Factorization
- Topic coherence
- Topic interpretation

Chapter 120: Sequence Models

- Sequential data
- Recurrent Neural Networks
- Hidden state
- Backpropagation Through Time
- Vanishing gradients
- LSTM
- GRU
- Bidirectional RNN
- Encoder-decoder
- Beam search

Phase 19: Transformers and LLMs

Chapter 121: Attention Mechanism

- Limitations of RNNs
- Encoder-decoder attention
- Query
- Key
- Value
- Attention scores
- Scaled dot-product attention
- Self-attention
- Cross-attention

- Multi-head attention

Chapter 122: Transformer Architecture

- Token embeddings
- Positional encoding
- Encoder blocks
- Decoder blocks
- Feed-forward networks
- Residual connections
- Layer normalization
- Attention masks
- Causal masks
- Encoder-decoder attention

Chapter 123: Tokenization

- Word-level tokenization
- Character tokenization
- Subword tokenization
- Byte Pair Encoding
- WordPiece
- SentencePiece
- Byte-level tokenization
- Vocabulary size
- Special tokens
- Context length

Chapter 124: Pretrained Language Models

- Language modelling
- Masked language modelling
- Causal language modelling
- BERT
- RoBERTa
- GPT
- T5
- Encoder-only models
- Decoder-only models
- Encoder-decoder models

Chapter 125: LLM Training

- Data collection

- Data filtering
- Pretraining
- Next-token prediction
- Supervised fine-tuning
- Instruction tuning
- Preference data
- RLHF
- Reward models
- DPO
- Model alignment

Chapter 126: Parameter-Efficient Fine-Tuning

- Full fine-tuning
- Feature extraction
- Adapter layers
- Prompt tuning
- Prefix tuning
- LoRA
- QLoRA
- Quantization
- Model distillation
- Catastrophic forgetting

Phase 20: Generative AI and Prompt Engineering

Chapter 127: Generative AI Fundamentals

- Generative vs discriminative models
- Autoregressive generation
- Tokens and context
- Temperature
- Top-K sampling
- Top-P sampling
- Maximum tokens
- Stop sequences
- Frequency penalties
- Hallucination
- Grounding

Chapter 128: Prompt Engineering

- System prompts

- User prompts
- Assistant messages
- Zero-shot prompting
- One-shot prompting
- Few-shot prompting
- Role prompting
- Contextual prompting
- Structured-output prompting
- Prompt templates
- Task decomposition
- ReAct prompting
- Tool-use prompting
- Prompt evaluation

Chapter 129: Autoencoders and Variational Autoencoders

- Encoder
- Bottleneck
- Decoder
- Reconstruction loss
- Denoising autoencoders
- Sparse autoencoders
- Variational autoencoders
- Latent space
- Reparameterization
- Anomaly detection

Chapter 130: Generative Adversarial Networks

- Generator
- Discriminator
- Adversarial training
- GAN loss
- Training instability
- Mode collapse
- DCGAN
- Conditional GAN
- CycleGAN
- StyleGAN

Chapter 131: Diffusion Models

- Forward diffusion
- Noise scheduling

- Reverse diffusion
- Noise prediction
- U-Net backbone
- Text conditioning
- Classifier-free guidance
- Latent diffusion
- Image generation
- Image editing

Phase 21: Embeddings, Vector Databases and RAG

Chapter 132: Embeddings

- Text embeddings
- Image embeddings
- Audio embeddings
- Multimodal embeddings
- Embedding dimensions
- Semantic similarity
- Normalization
- Cosine similarity
- Dot-product similarity
- Euclidean distance
- Embedding selection

Chapter 133: Vector Databases

- Vector storage
- Similarity search
- Exact nearest-neighbour search
- Approximate nearest neighbours
- HNSW
- IVF
- Product quantization
- Metadata
- Metadata filters
- Collections
- Namespaces
- FAISS
- Chroma
- Pinecone
- Weaviate

- Qdrant
- Milvus

Chapter 134: RAG Fundamentals

- Purpose of RAG
- Document ingestion
- Document loaders
- Text extraction
- Cleaning
- Chunking
- Chunk size
- Chunk overlap
- Embedding creation
- Vector indexing
- Query embeddings
- Similarity search
- Top-K retrieval
- Context construction
- Grounded generation
- Citations

Chapter 135: Advanced RAG

- Recursive chunking
- Semantic chunking
- Query rewriting
- Query expansion
- Multi-query retrieval
- Dense retrieval
- Sparse retrieval
- BM25
- Hybrid retrieval
- Reciprocal Rank Fusion
- Reranking
- Cross-encoders
- Parent-document retrieval
- Contextual compression

Chapter 136: Specialized RAG Systems

- Conversational RAG
- Multimodal RAG
- SQL RAG

- Knowledge-graph RAG
- Graph RAG
- Corrective RAG
- Self-RAG
- Agentic RAG
- Web RAG
- Multi-document RAG
- RAG caching
- RAG security

Chapter 137: RAG Evaluation

Retrieval evaluation

- Precision@K
- Recall@K
- Hit rate
- Mean Reciprocal Rank
- NDCG
- Context precision
- Context recall

Answer evaluation

- Answer relevance
- Faithfulness
- Groundedness
- Correctness
- Completeness
- Citation accuracy
- Hallucination rate
- Human evaluation
- LLM-as-a-judge
- RAGAS

Phase 22: AI Agents

Chapter 138: Agent Fundamentals

- Agent
- Goal
- Environment
- Observation

- Reasoning
- Planning
- Action
- Tools
- Memory
- Reflection
- Agent loop
- Workflow vs agent

Chapter 139: Tool-Using Agents

- Function calling
- Tool schemas
- Tool selection
- Tool execution
- Structured outputs
- API tools
- Search tools
- Database tools
- Code-execution tools
- Permission handling
- Error recovery

Chapter 140: Agent Architectures

- ReAct agents
- Plan-and-execute
- Router agents
- Supervisor agents
- Worker agents
- Multi-agent systems
- Hierarchical agents
- Human-in-the-loop
- Deterministic agent workflows

Chapter 141: Agent Memory and Evaluation

- Conversation memory
- Short-term memory
- Long-term memory
- Episodic memory
- Semantic memory
- Memory retrieval
- Task-success evaluation

- Tool-call accuracy
- Planning quality
- Safety evaluation
- Cost and latency evaluation

Chapter 142: Agent Frameworks

- LangChain
- LangGraph
- LlamaIndex
- AutoGen concepts
- Crew-based agent concepts
- Model Context Protocol
- Agent state management
- Checkpointing
- Observability
- Guardrails

Phase 23: Reinforcement Learning

Chapter 143: RL Fundamentals

- Agent
- Environment
- State
- Action
- Reward
- Policy
- Episode
- Return
- Discount factor
- Markov Decision Process
- Exploration vs exploitation

Chapter 144: Value-Based Learning

- State-value function
- Action-value function
- Bellman equation
- Dynamic programming
- Monte Carlo methods
- Temporal-difference learning
- Q-learning

- SARSA
- Epsilon-greedy strategy

Chapter 145: Deep Reinforcement Learning

- Deep Q-Network
- Experience replay
- Target networks
- Double DQN
- Dueling DQN
- Policy gradients
- Actor-Critic
- A2C
- A3C
- PPO
- Reward shaping
- Offline RL

Phase 24: Deployment and API Development

Chapter 146: Model Serialization

- Pickle
- Joblib
- Saving models
- Loading models
- Saving preprocessing pipelines
- Model compatibility
- Model versioning
- Safe serialization

Chapter 147: FastAPI and Flask

- REST API concepts
- HTTP methods
- JSON
- Request and response
- Status codes
- FastAPI
- Pydantic
- Route creation
- Input validation
- Exception handling

- CORS
- Swagger documentation
- Flask fundamentals
- API testing

Chapter 148: Streamlit and Gradio

- Application layout
- Widgets
- Forms
- File uploads
- Predictions
- Charts
- Session state
- Model caching
- Multipage applications
- User validation
- Error handling
- Deployment

Chapter 149: Docker

- Container concepts
- Image
- Container
- Dockerfile
- Build
- Run
- Ports
- Volumes
- Environment variables
- .dockerignore
- Docker Compose
- Multi-stage builds
- Container registries

Chapter 150: Cloud Deployment

- Cloud fundamentals
- Compute
- Storage
- Networking
- Virtual machines
- Serverless services

- AWS
- Azure
- Google Cloud
- Render
- CPU vs GPU deployment
- Secrets management
- Scaling
- Load balancing
- Logging

Phase 25: MLOps

Chapter 151: MLOps Fundamentals

- Development vs production ML
- Reproducibility
- Data versioning
- Code versioning
- Model versioning
- Experiment tracking
- Model registry
- Training pipelines
- Deployment pipelines
- Feature stores
- Metadata management

Chapter 152: MLOps Tools

- Git and GitHub
- DVC
- MLflow
- Weights & Biases
- Airflow
- Prefect
- Kubeflow
- Docker
- Kubernetes
- GitHub Actions
- CI/CD/CT

Chapter 153: Model Monitoring

- Model-performance monitoring

- Data-quality monitoring
- Input-schema validation
- Data drift
- Feature drift
- Prediction drift
- Concept drift
- Bias monitoring
- Latency
- Throughput
- Error rates
- Alerting
- Retraining triggers

Chapter 154: Production Deployment Strategies

- Batch inference
- Online inference
- Streaming inference
- Shadow deployment
- Canary deployment
- Blue-green deployment
- A/B testing
- Champion-challenger testing
- Rollback
- Autoscaling
- Caching

Phase 26: Data Engineering and Big Data

Chapter 155: Data Engineering Fundamentals

- ETL
- ELT
- Data pipelines
- Data warehouses
- Data lakes
- Data lakehouses
- Data marts
- Batch processing
- Stream processing
- Data quality

- Data lineage

Chapter 156: Big-Data Technologies

- Hadoop concepts
- HDFS
- MapReduce
- Apache Spark
- Spark DataFrames
- Spark SQL
- PySpark
- Kafka concepts
- Distributed computing
- Partitioning
- Fault tolerance

Phase 27: Responsible AI and Security

Chapter 157: Responsible AI

- Fairness
- Bias
- Transparency
- Explainability
- Accountability
- Privacy
- Consent
- Human oversight
- Accessibility
- Data governance
- Model cards
- Dataset documentation

Chapter 158: AI Fairness and Bias

- Historical bias
- Sampling bias
- Measurement bias
- Label bias
- Representation bias
- Selection bias
- Fairness metrics
- Demographic parity

- Equal opportunity
- Bias mitigation

Chapter 159: AI and LLM Security

- Adversarial examples
- Data poisoning
- Model extraction
- Model inversion
- Membership inference
- Prompt injection
- Indirect prompt injection
- Jailbreaking
- Sensitive-data leakage
- RAG poisoning
- Insecure tool use
- Access control
- Input validation
- Output filtering
- Rate limiting
- Audit logging

Phase 28: Advanced AI/ML Topics

Chapter 160: Graph Machine Learning

- Graphs
- Nodes and edges
- Adjacency matrices
- Degree
- Centrality
- Community detection
- Node embeddings
- Graph Convolutional Networks
- Graph Attention Networks
- Link prediction
- Knowledge graphs

Chapter 161: Causal Machine Learning

- Correlation vs causation
- Confounding
- Treatment and outcome

- Potential outcomes
- Average Treatment Effect
- Randomized experiments
- Propensity scores
- Difference-in-differences
- Instrumental variables
- Causal graphs
- Uplift modelling

Chapter 162: Multimodal AI

- Text-image models
- Vision-language models
- Audio-language models
- Image captioning
- Visual question answering
- Speech-to-text
- Text-to-speech
- Cross-modal embeddings
- Multimodal RAG
- Multimodal agents

Chapter 163: Edge AI and Model Optimization

- Model pruning
- Quantization
- Knowledge distillation
- Model compression
- ONNX
- TensorRT
- TensorFlow Lite
- Mobile deployment
- Edge-device constraints
- Memory optimization
- Latency optimization

Chapter 164: AutoML

- Automated preprocessing
- Automated feature engineering
- Automated model selection
- Automated tuning
- Neural architecture search
- Auto-sklearn

- H2O AutoML
- TPOT
- AutoKeras
- AutoML limitations

Phase 29: Software Engineering for AI/ML

Chapter 165: Git and GitHub

- Repository
- Clone
- Add
- Commit
- Push
- Pull
- Branch
- Merge
- Merge conflicts
- Pull requests
- .gitignore
- Tags and releases
- Code reviews
- GitHub Actions

Chapter 166: Clean ML Project Development

- Modular programming
- Separation of concerns
- Configuration files
- Environment variables
- Logging
- Type hints
- Docstrings
- Unit tests
- Integration tests
- Data tests
- Model tests
- Dependency management
- Packaging

- Reproducibility

Chapter 167: ML System Design

- Requirement clarification
- Data sources
- Label creation
- Feature pipelines
- Training architecture
- Model registry
- Batch vs online serving
- Real-time inference
- Scalability
- Latency
- Availability
- Monitoring
- Retraining
- Security
- Cost optimization

Phase 30: Projects and Interview Preparation

Chapter 168: Beginner Projects

- Student-marks prediction
- House-price prediction
- Salary prediction
- Iris classification
- Spam classification
- Sales analysis
- Customer segmentation
- Movie recommendation

Chapter 169: Intermediate Projects

- Customer-churn prediction
- Loan-approval prediction
- Credit-risk prediction
- Fraud detection
- Employee-attrition prediction
- Crop-yield prediction
- Retail analytics dashboard
- Sentiment analysis

- Sales forecasting
- Disease-risk prediction

Chapter 170: Advanced Projects

- Production RAG chatbot
- Multimodal RAG assistant
- AI customer-support agent
- SQL analytics agent
- Real-time fraud-detection system
- Recommendation engine
- Object-detection application
- Medical-image classifier
- End-to-end MLOps platform
- Model-monitoring dashboard

Chapter 171: Python Interview Preparation

- Mutable vs immutable
- List vs tuple
- Set vs dictionary
- Shallow vs deep copy
- Iterators vs generators
- Decorators
- Closures
- OOP
- Exception handling
- Memory management
- GIL
- Threading vs multiprocessing
- Python coding questions

Chapter 172: Machine-Learning Interview Preparation

- Bias vs variance
- Underfitting vs overfitting
- L1 vs L2
- Bagging vs boosting
- Regression vs classification
- Feature selection
- Cross-validation
- Data leakage
- Class imbalance
- Evaluation metrics

- Tuning
- Explainability
- Drift and monitoring

Chapter 173: Project Interview Preparation

Explain every project in this order:

1. Problem statement
2. Business importance
3. Dataset and features
4. Data cleaning
5. EDA findings
6. Feature engineering
7. Algorithms compared
8. Evaluation metrics
9. Final results
10. Deployment
11. Challenges
12. Future improvements

Recommended Learning Sequence

1. Python programming
2. NumPy, Pandas and visualization
3. SQL and databases
4. Mathematics and statistics
5. EDA and preprocessing
6. Classical machine learning
7. Evaluation and tuning
8. Unsupervised learning
9. Time series and recommendations
10. Deep learning
11. Computer vision
12. NLP
13. Transformers and LLMs
14. Generative AI and prompt engineering
15. Embeddings, vector databases and RAG
16. AI agents
17. Deployment and APIs
18. Docker and cloud
19. MLOps and monitoring

20. Responsible AI and security
21. Advanced projects
22. Interview preparation

This master syllabus contains **173 chapters**, covering the complete path from Python fundamentals to production-level AI/ML, Generative AI, RAG, agents and MLOps.